SIMATS ENGINEERING
Department of Electrical and Electronics Engineering
ASSESSMENT TOOL –

Engineer to Excel

Course Code:	ECA1418	Course Title:	Embedded Systems
CO Assessed:	CO : 1	Assessment:	AT-1
Weightage:		Total Marks:	
CO1 Statement:			
Student Name:	M.Nandhini	Reg. No.:	192472318
Section / Batch:	3rd year (Slot-B)	Date:	24-07-26

Question Type	Description	Questions	Total Marks
	Concept Mapping	Q - Q	100

Code of Conduct

I M.Nandhini (192472318) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: M.Nandhini

RUBRICS FOR MARKS CALCULATION

Task	(%)	(%)	(%)	(%)	Total Marks (Each -)
					20
				Overall Total ( )	

Faculty In-charge	Course Coordinator	Head of Department
Name: Signature:	Signature:	Signature:
Date: 24/7/26	Date:	Date:

M. Nandhini (192472318)
ECA1418 - Embedded Systems
Dr. C.M. ARUN KUMAR
Slot - B

Concept Mapping

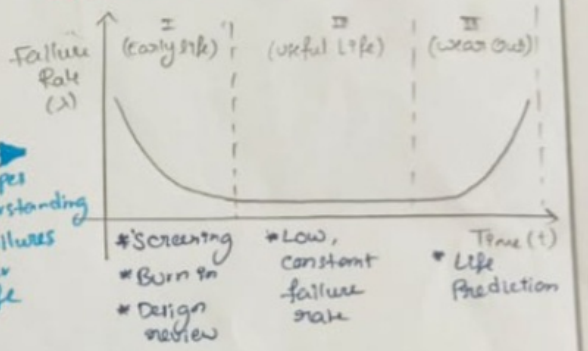
Embedded Product Reliability

Goal
Ensure reliable and Safe Operation of Embedded Products throughout their lifecycle.

- Metrics derived from models
- Shapes understanding of failures over life

Basic Reliability Metrics
- Failure Rate (λ)
- MTTF (Mean Time To Failure)
  $MTTF = 1/\lambda$
- Reliability Function ($R(t)$)

Bathtub Curve (Failure Rate vs Time)



Reliability Growth
- Improve reliability through testing, debugging, design changes
- Track metrics (failure rate, MTBF)
- Aim: Move Product early to "Useful Life" region.

Reliability Models

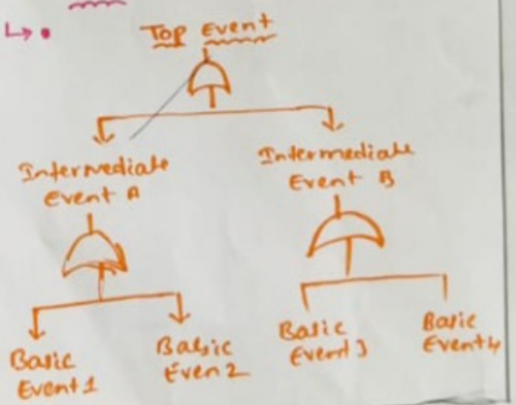
Exponential Model	Weibull Model
$R(t) = e^{-\lambda t}$	$R(t) = e^{-(\frac{t}{\eta})^{\beta}}$
$MTTF = 1/\lambda$	$\beta < 1$: Decreasing
	$\beta = 1$: Constant
	$\beta > 1$: Increasing

Redundancy Strategies
- Hardware Redundancy
  - Component duplication
  - N-modular redundancy
- Information Redundancy
  - Error detection
- Time Redundancy
  - watchdogs, retries
- Diversity
  - Different technology to avoid common cause failure.

Reliability Analysis & Risk Identification Tools

FMEA
- Failure Mode and Effects Analysis
- Systematic bottom-up analysis to identify Potential failure models and their impact
  - Step
  - focus
  - output

FTA



Influence on Designs Decisions Across the Embedded Product Development lifecycle
- Requirements & Concept
- Architecture & Design
- Implementation
- verification & validation
- Production & Deployment
- Operation & Maintance
- END of LIFE & Feedback